CN Practical 14: Create Tree Topology Using Cisco Packet Tracer

Aim: To create a tree topology network using Cisco Packet Tracer and test communication between connected computers.

Theory:

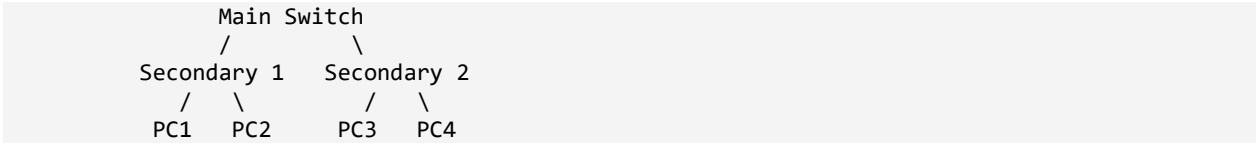
Tree topology is a hierarchical network arrangement in which devices are connected in multiple levels. It combines characteristics of star topology with a hierarchical backbone structure. A typical tree network consists of a main switch or root device connected to secondary switches, which then connect to computers and other end devices.

Tree topology is suitable for schools, colleges, offices and large organisations where the network must be divided into separate sections or departments. It makes the network easier to expand and organise. However, failure of a main backbone link or central device can affect the devices connected below that point. The design also requires careful planning of IP addresses and network connections.

Software and materials required:

- Cisco Packet Tracer
- One main switch
- Two secondary switches
- Four PCs
- Ethernet connections

Network diagram:



IP addressing table:

Device	IP address	Subnet mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0
PC4	192.168.1.4	255.255.255.0

Procedure:

1. Open Cisco Packet Tracer on the computer.
2. Select and place one main switch in the workspace.
3. Add two secondary switches and four PCs.
4. Connect the main switch to both secondary switches using suitable Ethernet links.
5. Connect PC1 and PC2 to the first secondary switch.
6. Connect PC3 and PC4 to the second secondary switch.
7. Open the IP Configuration window of each PC.
8. Assign the IP addresses shown in the table and use the subnet mask 255.255.255.0.

9. Wait until the network links become active.

10. Open the command prompt on PC1.

11. Execute ping 192.168.1.4 to test connectivity with PC4.

12. Observe the results and verify that the devices can communicate.

Observation:

The network was arranged in a hierarchical structure, with the main switch connected to two secondary switches. Each secondary switch connected to two PCs. The computers were assigned IP addresses in the same subnet, and the ping command was used to test communication between devices on different branches of the tree.

Conclusion:

A tree topology network was created using Cisco Packet Tracer. The practical demonstrated hierarchical network design, device interconnection and IP address configuration. Communication between computers was tested successfully using the ping command.